Optimization Methods and Software

ISSN: 1055-6788 (Print) 1029-4937 (Online) Journal homepage: www.tandfonline.com/journals/goms20

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To cite this article: Mark Sofroniou & Giulia Spaletta (2005) Derivation of symmetric composition constants for symmetric integrators, Optimization Methods and Software, 20:4-5, 597-613, DOI: 10.1080/10556780500140664

To link to this article: https://doi.org/10.1080/10556780500140664

Published online: 15 Aug 2006.

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Optimization Methods and Software
Vol. 20, Nos. 4–5, August–October 2005, 597–613

Derivation of symmetric composition constants for symmetric integrators

MARK SOFRONIOU† and GIULIA SPALETTA*‡

†Wolfram Research, Champaign, IL, USA
‡Mathematics Department, Bologna University, Italy

(Received 6 November 2003; in final form 30 June 2004)

This work focuses on the derivation of composition methods for the numerical integration of ordinary differential equations, which give rise to very challenging optimization problems. Composition is a useful technique for constructing high order approximations, while conserving certain geometric properties. We survey existing composition methods and describe results of an intensive numerical search for new methods. Details of the search procedure are given along with numerical examples, which indicate that the new methods perform better than previously known methods. Some insight into the location of global minima for these problems is obtained as a result.

Keywords: Geometric numerical integration; Composition methods; Splitting methods; Symplectic methods; Computer algebra systems

2000 Mathematics Subject Classifications: 65L05; 65P10; 68W30

1. Introduction

Composition is a useful device for raising the order of a numerical integration scheme. In contrast to the Aitken–Neville algorithm used in extrapolation, composition can conserve desirable geometric properties of a base integration method, such as symplecticity [see for example ref. 1]. One motivation for this work was to review the best available composition methods in order to include them in a new differential equation solver being developed in *Mathematica*. We also wanted to investigate the derivation of high order methods because of the relatively large error constants that were observed for a method of order ten in ref. [2].

The numerical search procedure that has been adopted makes use of new optimization capabilities that are available in *Mathematica* – results presented here were obtained using version 5.0.1. Of particular importance is substantially improved linear algebra. Another feature avoids the need to continually process input expressions and is particularly advantageous for a randomized search. An additional advantage is the automation of the computation of

*Corresponding author. Email: giulia@cs.unibo.it

Optimization Methods and Software
ISSN 1055-6788 print/ISSN 1029-4937 online © 2005 Taylor & Francis Group Ltd
http://www.tandf.co.uk/journals
DOI: 10.1080/10556780500140664

exact expressions for derivatives – these are computed lazily when needed and cached for efficient recall. Using exact expressions for derivatives in optimization routines is often more efficient and gives more accurate results than numerical differences [3].

The numerical optimization search can be very intensive, especially when deriving high order composition schemes. Therefore, the high level parallel capabilities of *gridMathematica* have been used in conjunction with the parallel computing toolkit **Parallel.m** [4]. All computations were carried out on a dedicated grid of eight dual-PowerPC G4 processor Apple Macintosh OSX Xserve machines. Linear algebra is performed using the LAPACK library [5] using optimized BLAS routines developed by Apple specifically for the G4 processor.

This article is organized as follows. In section 2, composition methods are introduced together with the algebraic conditions, which determine the order. In section 3, details of the numerical optimization process are given. In section 4, new methods that have been derived are presented and compared to existing methods. In section 5, the results of some numerical experiments are given to compare the new methods with existing methods.

2. Background

Consider an initial value problem: $\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y})$, $\mathbf{f}$: $\mathbb{R} \times \mathbb{R}^d \mapsto \mathbb{R}^d$ and $\mathbf{y}(0) = \mathbf{y}_0$ given. An s-stage composition method $\Psi_{\mathbf{f},h}$ is given by:

$$\Psi_{\mathbf{f},h} = \Phi^{(1)}_{\mathbf{f},\gamma_s h} \circ \cdots \circ \Phi^{(s)}_{\mathbf{f},\gamma_1 h}, \tag{1}$$

where each $\Phi^{(i)}_{\mathbf{f},\gamma_i h}$ is a basic integration method for $\mathbf{f}$, that takes a step of size $\gamma_i h$, with γ_i and h given real numbers. Often, we are interested in composition methods which involve the same order p base method with different step sizes:

$$\Psi_{\mathbf{f},h} = \Phi_{\mathbf{f},\gamma_s h} \circ \cdots \circ \Phi_{\mathbf{f},\gamma_1 h}. \tag{2}$$

An interesting special case of (2) is *symmetric composition*, obtained by taking:

$$\gamma_i = \gamma_{s-i+1}, \quad i = 1, \ldots, \left\lfloor \frac{s}{2} \right\rfloor. \tag{3}$$

We will use $\boldsymbol{\gamma}$ to denote the vector $[\gamma_1, \ldots, \gamma_s]^{\mathrm{T}}$.

There exists a general composition technique that can be used to raise the order [6,7]. For even order p, the composition (2) has order at least $p + 1$ if:

$$\sum_{i=1}^{s} \gamma_i = 1, \quad \sum_{i=1}^{s} \gamma_i^{p+1} = 0. \tag{4}$$

Let $\Phi^{2k}_{\mathbf{f}}$ denote a symmetric method of order $2k$. Then, a symmetric method of order $2k + 2$ can be obtained as:

$$\Phi^{2k+2}_{\mathbf{f},h} = \left(\Phi^{2k}_{\mathbf{f},\alpha h}\right)^n \left(\Phi^{2k}_{\mathbf{f},\beta h}\right)^m \left(\Phi^{2k}_{\mathbf{f},\alpha h}\right)^n, \tag{5}$$

where $\alpha = 1/(2n - (2nm^{2k})^{1/(2k+1)})$ and $\beta = (1 - 2n\alpha)/m$ are real values and n, m are positive integers which determine the number of stages (ref. [8] for a recent summary). In this way, an increase in order of two is achieved for (at least) three times the cost of the base method. This turns out to be asymptotically larger than the growth of the number of stages in a direct construction.

In the sequel, we will restrict our attention to methods of the form (2), subject to the constraint (3), and where the integration method is symmetric of order $p = 2$. It is worth mentioning that composition of a method with its adjoint is another interesting case of (1) since it allows an order increase also for odd order. The methods derived here can also be used in this context (see ref. [1] for more details).

Order conditions for the direct construction of composition methods can be derived in several ways (see ref. [1,8] for a summary). An elegant graph theoretical approach, extending the rooted tree formalism used in the derivation of Runge–Kutta methods, has been developed in ref. [9] and is succinctly reviewed in ref. [1]. The order conditions for symmetric composition of symmetric methods through order eight (as well as the trees required for order ten) are listed in ref. [1]. In order to make our description self-contained, we list the order conditions through order ten.

$$\sum_{k=1}^{s} \gamma_k = 1, \tag{6}$$

$$\sum_{k=1}^{s} \gamma_k^3 = 0, \tag{7}$$

$$\sum_{k=1}^{s} \gamma_k^5 = 0, \quad \sum_{k=1}^{s} \gamma_k^3 \left(\sum_{l=1}^{k}{}' \gamma_l \right)^2 = 0, \tag{8}$$

$$\begin{gathered} \sum_{k=1}^{s} \gamma_k^7 = 0, \quad \sum_{k=1}^{s} \gamma_k^5 \left(\sum_{l=1}^{k}{}' \gamma_l \right)^2 = 0, \\ \sum_{k=1}^{s} \gamma_k^3 \sum_{l=1}^{k}{}' \gamma_l \sum_{m=1}^{k}{}' \gamma_m^3 = 0, \quad \sum_{k=1}^{s} \gamma_k^3 \left(\sum_{l=1}^{k}{}' \gamma_l \right)^4 = 0, \end{gathered} \tag{9}$$

$$\begin{gathered} \sum_{k=1}^{s} \gamma_k^9 = 0, \quad \sum_{k=1}^{s} \gamma_k^7 \left(\sum_{l=1}^{k}{}' \gamma_l \right)^2 = 0, \\ \sum_{k=1}^{s} \gamma_k^5 \sum_{l=1}^{k}{}' \gamma_l \sum_{m=1}^{k}{}' \gamma_m^3 = 0, \quad \sum_{k=1}^{s} \gamma_k^3 \sum_{l=1}^{k}{}' \gamma_l \sum_{m=1}^{k}{}' \gamma_m^5 = 0, \\ \sum_{k=1}^{s} \gamma_k^3 \left(\sum_{l=1}^{k}{}' \gamma_l \right)^2 \sum_{m=1}^{k}{}' \gamma_m^3 \sum_{n=1}^{m}{}' \gamma_n = 0, \quad \sum_{k=1}^{s} \gamma_k^5 \left(\sum_{l=1}^{k}{}' \gamma_l \right)^4 = 0, \\ \sum_{k=1}^{s} \gamma_k^3 \left(\sum_{l=1}^{k}{}' \gamma_l \right)^3 \sum_{m=1}^{k}{}' \gamma_m^3 = 0, \quad \sum_{k=1}^{s} \gamma_k^3 \left(\sum_{l=1}^{k}{}' \gamma_l \right)^6 = 0. \end{gathered} \tag{10}$$

Summation with a prime indicates that the last term γ_i^j is taken as $\gamma_i^j/2$.

3. Optimization details

The minimal number of stages for an order $p = 2k$, $p \leq 10$, composition scheme subject to the constraint (3) is $2^k - 1$. When the number of stages is minimal, the number of free parameters and algebraic constraints is the same and Newton's method can be used to find

solutions using randomized initial search data. However, a number of numerical comparisons of composition methods have shown that it is preferable to consider $2^k + 1$ stages and use the additional free parameter to optimize the coefficients in some way [1,10]. The main suggested choices are as follows.

- Minimize some norm of the principal truncation error [11]. Performance is very problem dependent (see for example [1]).
- Minimize $\|\boldsymbol{\gamma}\|_1$, which corresponds to the total distance travelled in the integration interval.
- Minimize $\|\boldsymbol{\gamma}\|_\infty$, which corresponds to the largest intermediate step size.

We have chosen to minimize the one-norm because of its physical significance and because it readily allows us to test our derivation framework with some of the best known existing methods. Furthermore, minimizing the one-norm also reduces cumulative roundoff error which is important in geometric integration and particularly of concern in the methods of interest here, since a large number of stages are involved.

For $x \in \mathbb{R}$, it holds that $|x| = \sqrt{x^2}$ which gives a form that can be differentiated symbolically and can therefore be used in an objective function for a conventional minimization procedure. The singularity in the derivative of the objective function at zero did not give rise to numerical difficulties, since it did not prevent the sign of components from changing during the optimization process. It is known that some of the compositions weights are necessarily negative. As well as reducing the magnitude of the error coefficients, additional stages give smaller values which reduce the effect of cancellation and cumulative rounding errors. Recent work has indicated that it might be fruitful to search for composition methods with yet additional stages [12,13]. Therefore, we also performed a search for such methods.

A nonlinear constrained optimization problem can be converted into an unconstrained least squares problem by the construction of a suitable *penalty function* [14]. The problem of interest has the form of nonlinear parameter estimation and can be modelled as the minimization of a sum of squares of nonlinear functions: $\min_{\mathbf{x}} \sum_{i=1}^{r+a} f_i(\mathbf{x})^2/2,\ f_i\colon \mathbb{R}^r \mapsto \mathbb{R}$. In the overdetermined case, $a \geq 1$, this type of least squares optimization problem is well suited to the Levenberg–Marquardt method [3]. Success in using this type of approach for deriving coefficients for numerical methods for differential equations has been reported in ref. [15]. We use a new feature of the **Levenberg Marquart** method of **FindMinimum** in *Mathematica*, which allows the direct specification of a function to compute the residual vector $[f_1, \ldots, f_{r+a}]^{\mathrm{T}}$. Taking into account (3), we construct the residual vector from a vector of the order conditions augmented by the quantity:

$$\left[\sqrt{2}\sqrt[4]{\gamma_1^2}, \ldots, \sqrt{2}\sqrt[4]{\gamma_{\lfloor s/2\rfloor}^2}, \sqrt[4]{\gamma_{\lfloor s/2\rfloor+1}^2}\right]^{\mathrm{T}}.$$

This allows a search for composition methods with several additional stages, since the number of components of the residual vector exceeds the number of free parameters. Each order condition is scaled by a *penalty parameter* κ to ensure that the first few digits are a reasonable solution to the order conditions – we have found that $\kappa = 10^8$ works reasonably well in practice. For each random input vector, the local minimization is accepted if an absolute or relative error of 10^{-4} is achieved in the solution or if an absolute error of 10^{-4} is achieved in the objective function. The search is terminated for each random input if the maximum number of 10^4 iterations is exceeded or if a sufficient decrease in the merit function of the line search is not obtained. The search was terminated when all processors had run for a total of 10^6 random initial vectors (10^5 initial vectors was adequate for methods of order six and eight).

Obtaining good initial data for the numerical search is crucial for finding solutions, especially at high order. Despite a randomized contribution, we have therefore constructed data

that incorporates some structural knowledge about the problem. The random initial search data for $\boldsymbol{\gamma}$ uses values in $(0, c)$ or $(-c, 0)$, $c \in \mathbb{R}$, where negative intervals are generated roughly one in three times. The position of the negative weights is also randomized. The actual values found by the minimization process may exceed the initial values in magnitude and we found that $c = 2/3$ provides a reasonable initial bound. We discard an initial random vector (and begin generating a new one), if the partial sum of the components generated so far is either smaller than zero or larger than one in magnitude. Motivation for these criteria can be found in section 4.4. Symmetry is taken into account and the sum of the initial vector components is exactly one to guarantee that the data form at least an order one (consistent) integration method. Initializing the random seed for each processor to a different value ensures that distinct random sequences are generated. To incorporate a measure of the best solution found so far into the random starting data, each processor stores the best value for the objective function. A new random vector is rejected if the total distance (sum of absolute values of its components) exceeds the best value found so far. This idea is based on the clustering technique that is common in global optimization. The random vector may not be close to a solution of the order conditions (in which case the maximum number of iterations will be reached), but it helps to avoid spending time generating solutions with a larger norm value than the best obtained so far in the search.

The Levenberg–Marquardt procedure gives solutions that are only accurate in the first few digits. Starting from this initial approximation, refinement of the coefficients can be performed using Newton's method on the Lagrange multiplier equations, that are formed from $\|\boldsymbol{\gamma}\|_1$ and the order conditions. This ensures that the digits of the method correspond to those of a local minimum. For example, consider a 5 stage method of order four subject to the constraint (3) which minimizes $\|\boldsymbol{\gamma}\|_1$. The Lagrange multiplier equations are obtained by considering the conditions for a minimum of the Lagrangian function:

$$\begin{aligned} L(\gamma_1, \gamma_2, \gamma_3, \lambda_1, \lambda_2) = {} & 2\sqrt{\gamma_1^2} + 2\sqrt{\gamma_2^2} + \sqrt{\gamma_3^2} - \lambda_1(2\gamma_1 + 2\gamma_2 + \gamma_3 - 1) \\ & - \lambda_2(2\gamma_1^3 + 2\gamma_2^3 + \gamma_3^3) \end{aligned}$$

Initial starting values for the Lagrange multipliers λ_i in the Newton procedure are obtained by first solving a linear subsystem of the Lagrange multiplier equations, using the initial approximations for γ_i.

4. Methods

We now present a number of composition methods found by our numerical search. High order schemes are sometimes implemented using in extended precision (see [1] and the experiments in section 5) so we have refined to 32 decimal digits.

4.1 *Order four*

Methods of order four can be found in closed form. The three stage method of Yoshida [7] corresponds to the composition (5) with $k = m = n = 1$. The five stage method of Suzuki [6] corresponds to the composition (5) with $k = m = 1$, $n = 2$. The method of Suzuki often performs better than the method of Yoshida because the smaller error constants mean that larger step sizes can be taken, which offsets the larger number of stages (see for example [1]). It is known that Suzuki's method is optimal for $\|\boldsymbol{\gamma}\|_1$ and $\|\boldsymbol{\gamma}\|_\infty$ [1,10].

4.2 *Order six*

Consider the order conditions (6–8) subject to the constraint (3). Yoshida's method D [7] is the optimal solution with 7 stages and a closed form solution is given by Koseleff in ref. [16]. The optimal 9 stage method that we found coincides with s9odr6b of Kahan and Li [10]. The best 11 stage method that we found has coefficients:

$$\begin{aligned}
\gamma_1 = \gamma_{11} &= 0.21375583945878254555518066964857\\
\gamma_2 = \gamma_{10} &= 0.18329381407425713911385974425217\\
\gamma_3 = \gamma_9 &= 0.17692819473098943794898811709929\\
\gamma_4 = \gamma_8 &= -0.44329082681170215849622829626258\\
\gamma_5 = \gamma_7 &= 0.11728560432865935385403585669136\\
\gamma_6 &= 0.50405474843802736404832781714239
\end{aligned} \tag{11}$$

The best 13 stage method that we found has coefficients:

$$\begin{aligned}
\gamma_1 = \gamma_{13} &= 0.13861930854051695245808013042625\\
\gamma_2 = \gamma_{12} &= 0.13346562851074760407046858832209\\
\gamma_3 = \gamma_{11} &= 0.13070531011449225190542755785015\\
\gamma_4 = \gamma_{10} &= 0.12961893756907034772505366537091\\
\gamma_5 = \gamma_9 &= -0.35000324893920896516170830911323\\
\gamma_6 = \gamma_8 &= 0.11805530653002387170273438954049\\
\gamma_7 &= 0.39907751534871587459988795520665
\end{aligned} \tag{12}$$

It can be observed from table 1 that both methods (11) and (12) improve on method s9odr6b of ref. [10] at the cost of additional stages.

Table 1. Comparison of norms of coefficients for various composition methods.

Order	Stages	Method	$\lVert\boldsymbol{\gamma}\rVert_1$	$\lVert\boldsymbol{\gamma}\rVert_\infty$
4	3	(5) $k = 1, m = 1, n = 1$	4.40483	1.702414
4	5	(5) $k = 1, m = 1, n = 2$	2.31593	0.657963
6	7	Method D of ref. [7]	5.71072	1.315186
6	9	s9odr6b of ref. [10]	3.82491	0.798564
6	11	Method (11)	2.77316	0.504055
6	13	Method (12)	2.40001	0.399078
8	15	Refs. [11, 17]	6.52563	0.796888
8	17	s17odr8b of ref. [10]	5.34955	0.606074
8	19	Method (13)	4.42210	0.494122
8	21	Method (14)	3.67674	0.382636
10	31	Method (15)	7.54471	0.607728
10	33	s33odr10a of ref. [10]	13.6857	0.955298
10	33	s33odr10c of ref. [10]	12.3702	0.836472
10	33	(3.15) of ref. [1]	8.88706	0.535918
10	33	Method (16)	6.79002	0.486326
10	35	Method (17)	6.18133	0.411431

4.3 *Order eight*

Consider the conditions (6–9) subject to the constraint (3). Methods with 15 stages are given by Yoshida in ref. [7]. We have verified that the optimal solution is that given in refs. [11,17]. The optimal 17 stage method that we found coincides with s17odr8b of Kahan and Li [10]. The best 19 stage method that we found has coefficients:

$$\begin{aligned}
\gamma_1 = \gamma_{19} &= 0.102369976919196772179472330167 68\\
\gamma_2 = \gamma_{18} &= 0.151937195421241500421225175198 86\\
\gamma_3 = \gamma_{17} &= -0.257585007988004197146753453204 34\\
\gamma_4 = \gamma_{16} &= 0.222072809073596277452873208241 57\\
\gamma_5 = \gamma_{15} &= 0.144280791092728571694099774000 23\\
\gamma_6 = \gamma_{14} &= 0.459024127914542530447389881945 08\\
\gamma_7 = \gamma_{13} &= -0.350879810350093462559038400933 39\\
\gamma_8 = \gamma_{12} &= 0.126329693889236743601578185621 48\\
\gamma_9 = \gamma_{11} &= 0.149511435687219886022490804434 02\\
\gamma_{10} &= -0.494122423319329244226675010942 40
\end{aligned} \tag{13}$$

The best 21 stage method that we found has coefficients:

$$\begin{aligned}
\gamma_1 = \gamma_{21} &= 0.106477289845500318239319678548 96\\
\gamma_2 = \gamma_{20} &= 0.108374086458357263974334105915 46\\
\gamma_3 = \gamma_{19} &= 0.353378210526543424195345413240 80\\
\gamma_4 = \gamma_{18} &= -0.233414140231650821987802811283 19\\
\gamma_5 = \gamma_{17} &= -0.244452667915288412694621714132 16\\
\gamma_6 = \gamma_{16} &= 0.113178484357556333147009525155 99\\
\gamma_7 = \gamma_{15} &= 0.118929056250003500626929722839 51\\
\gamma_8 = \gamma_{14} &= 0.126039123218259881403056702683 65\\
\gamma_9 = \gamma_{13} &= 0.125817187361760418043923916415 87\\
\gamma_{10} = \gamma_{12} &= 0.116991350192176421807228814335 33\\
\gamma_{11} &= -0.382635960126436653509446707440 40
\end{aligned} \tag{14}$$

It can be observed from table 1 that both methods (13) and (14) improve on the method s17odr8b of ref. [10] at the cost of additional stages.

4.4 *Order ten*

Kahan and Li derived the first direct methods of order ten [10]. Hairer *et al.* subsequently derived a better method [1]. Both schemes use 33 stages. Consider the conditions (6–10) subject to the constraint (3). For completeness, we present the best method with the minimal

number of 31 stages that was found in our search:

$$
\begin{aligned}
\gamma_1 = \gamma_{31} &= 0.14998070054317051502516939497857\\
\gamma_2 = \gamma_{30} &= 0.09120863510148929199610512151446 2\\
\gamma_3 = \gamma_{29} &= 0.50623124887796194535266557555255\\
\gamma_4 = \gamma_{28} &= 0.09478971592588915409423145408920 4\\
\gamma_5 = \gamma_{27} &= -0.19520875735034504160990960439871\\
\gamma_6 = \gamma_{26} &= -0.38816256756251756192331854792644\\
\gamma_7 = \gamma_{25} &= -0.27450555650873276528931810649505\\
\gamma_8 = \gamma_{24} &= 0.14264675556451861069659069043321\\
\gamma_9 = \gamma_{23} &= 0.06710251896682534934687739603780 9\\
\gamma_{10} = \gamma_{22} &= -0.19643186370792190448674783323248\\
\gamma_{11} = \gamma_{21} &= 0.29602854892160888804740587728740\\
\gamma_{12} = \gamma_{20} &= 0.18917810251470701571585847859316\\
\gamma_{13} = \gamma_{19} &= 0.19394700133244324371285167850479\\
\gamma_{14} = \gamma_{18} &= 0.10120067580762238380456506324802\\
\gamma_{15} = \gamma_{17} &= -0.58186926782264021140090352527182\\
\gamma_{16} &= 0.60772821879184217383575377417062
\end{aligned}
\tag{15}
$$

The best 33 stage method that we found has coefficients:

$$
\begin{aligned}
\gamma_1 = \gamma_{33} &= 0.07071126158608539907930277181020 3\\
\gamma_2 = \gamma_{32} &= 0.09034208093726756834557791438923 4\\
\gamma_3 = \gamma_{31} &= 0.14103133297152486103524322594476\\
\gamma_4 = \gamma_{30} &= 0.40206004554029767537357060971803\\
\gamma_5 = \gamma_{29} &= -0.30239722849131075165735249848238\\
\gamma_6 = \gamma_{28} &= -0.22462355658241460137093154363351\\
\gamma_7 = \gamma_{27} &= 0.06149698895606312194038070706841 1\\
\gamma_8 = \gamma_{26} &= 0.11346847775740802675296685287062\\
\gamma_9 = \gamma_{25} &= 0.23654672241381781124636015203490\\
\gamma_{10} = \gamma_{24} &= 0.27211409645898977643699556260890\\
\gamma_{11} = \gamma_{23} &= 0.07612941847027723438653090665102 4\\
\gamma_{12} = \gamma_{22} &= -0.18543093454238185309165565783301\\
\gamma_{13} = \gamma_{21} &= -0.46495110925607623804616342747217\\
\gamma_{14} = \gamma_{20} &= 0.10423014962104084592532590279051\\
\gamma_{15} = \gamma_{19} &= 0.13621181452383232935841998116651\\
\gamma_{16} = \gamma_{18} &= -0.27010275720513252644976102610064\\
\gamma_{17} &= 0.48632639368142264147037913293721
\end{aligned}
\tag{16}
$$

The best 35 stage method that we found has coefficients:

$$
\begin{aligned}
\gamma_1 = \gamma_{35} &= 0.0787957225216864192639076793376 84\\
\gamma_2 = \gamma_{34} &= 0.3130961034151085277648124719264 7\\
\gamma_3 = \gamma_{33} &= 0.0279183832350780661095202732752 99\\
\gamma_4 = \gamma_{32} &= -0.2295928415939070941512133967965 5\\
\gamma_5 = \gamma_{31} &= 0.1309620610771648631746568592796 1\\
\gamma_6 = \gamma_{30} &= -0.2697334056545107143446097322241 1\\
\gamma_7 = \gamma_{29} &= 0.0749733431558914356661371056414 10\\
\gamma_8 = \gamma_{28} &= 0.1119934239998102048895750807364 0\\
\gamma_9 = \gamma_{27} &= 0.3661334495462267511931481235315 0\\
\gamma_{10} = \gamma_{26} &= -0.3991056301360358978786298105834 0\\
\gamma_{11} = \gamma_{25} &= 0.1030873985274710773158027700137 2\\
\gamma_{12} = \gamma_{24} &= 0.4114308739558902378207041189760 8\\
\gamma_{13} = \gamma_{23} &= -0.0048663605831352617621956593099771\\
\gamma_{14} = \gamma_{22} &= -0.3920333537086399064480819364261 0\\
\gamma_{15} = \gamma_{21} &= 0.0519425029624496470371829040159 76\\
\gamma_{16} = \gamma_{20} &= 0.0506650907599244963358743441568 66\\
\gamma_{17} = \gamma_{19} &= 0.0496743706397298790545688002794 61\\
\gamma_{18} &= 0.0493177357595945379176800083393 38
\end{aligned}
\tag{17}
$$

From table 1, it can be seen that method (17) has the smallest norm values amongst tenth order methods and all of the new schemes have smaller one-norm values than the previously known methods.

A useful technique for graphical comparison of the coefficients is the *cumulative weight plot*, which plots the pairs (c_i, i) for $i = 1, \ldots, s$, where $c_i = \sum_{j=1}^{i} \gamma_j$. Figure 1 presents cumulative weight plots for several tenth order methods. The method (16) has the advantage that since $c_i \in [0, 1]$, all sample points lie in the current integration interval. The method s33odr10a of Kahan and Li [10] was constructed to have this property, but has much larger norm values (see table 1).

Figure 2 presents the cumulative weight plots for the new tenth order methods. Method (15) has sample points that lie outside of the integration interval. Method (17) has a significantly smaller deviation from an idealized straight line between $(0, 0)$ and $(1, s)$ than the other methods.

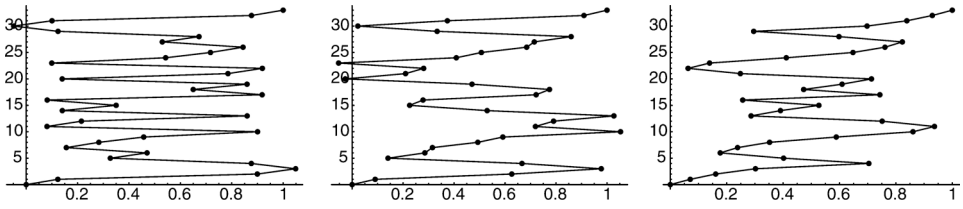

Figure 1. Cumulative weight plot for method s33odr10c of Kahan and Li [10] (left) the method (3.15) of Hairer *et al.* [1] (center) and the new method (16) (right).

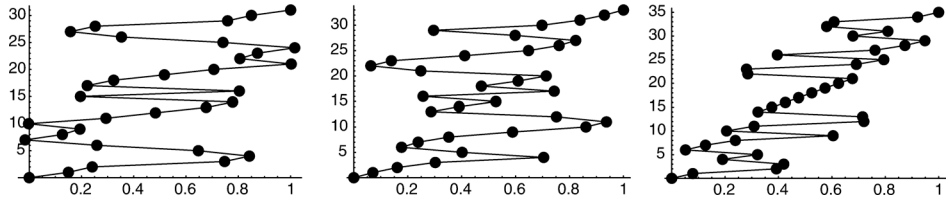

Figure 2. Cumulative weight plot for new tenth order methods with 31 stages (15) (left) 33 stages (16) (center) and 35 stages (17) (right).

5. Numerical experiments

To compare various schemes, we use the coefficient plug-in mechanism of a new framework for composed numerical integrators in *Mathematica* [18]. The large number of stages used by high order methods requires careful implementation to reduce cumulative roundoff error. An increment formulation coupled with compensated summation is used to reduce the effect of rounding errors (see ref. [2] for details). Arbitrary precision software arithmetic with 32 decimal digits of precision is used to obtain a smooth error-work comparison at high accuracy and all coefficients used the same level of precision. High precision reference solutions were computed using the method **Extrapolation**.

Splitting is a generalization of composition where the vector field is broken up in an additive fashion. Consider the case in which the vector field can be split into two parts $\mathbf{f} = \mathbf{f}_1 + \mathbf{f}_2$. Let $\varphi_{\mathbf{f}_1,h}$ and $\varphi_{\mathbf{f}_2,h}$ denote the exact solution (flow). A second order symmetric integration method can be constructed using a Strang splitting [19,20]:

$$\varphi_{\mathbf{f}_1,h/2} \circ \varphi_{\mathbf{f}_2,h} \circ \varphi_{\mathbf{f}_1,h/2}. \tag{18}$$

The splitting methods used for the experiments exploit the group property of flows:

$$\varphi_{\mathbf{f},h_1+h_2} = \varphi_{\mathbf{f},h_2} \circ \varphi_{\mathbf{f},h_1}. \tag{19}$$

For example, splitting into two vector fields yields methods of the form

$$\varphi_{\mathbf{f}_1,a_{s+1}} \circ \varphi_{\mathbf{f}_2,b_s} \circ \varphi_{\mathbf{f}_1,a_s} \circ \cdots \circ \varphi_{\mathbf{f}_2,b_1} \circ \varphi_{\mathbf{f}_1,a_1}. \tag{20}$$

Using the group property of flows, the coefficients for this splitting method are related to the composition coefficients in the following way:

$$\begin{aligned} a_1 &= \frac{\gamma_1}{2}, & b_1 &= \gamma_1, \\ a_2 &= \frac{(\gamma_1+\gamma_2)}{2}, & b_2 &= \gamma_2, \\ &\vdots & &\vdots \\ a_s &= \frac{\gamma_{s-1}+\gamma_s}{2}, & b_s &= \gamma_s, \\ a_{s+1} &= \frac{\gamma_s}{2}. \end{aligned}$$

By the First Same As Last (FSAL) property, the method (20) requires s evaluations of $\varphi_{\mathbf{f}_1}$ and $\varphi_{\mathbf{f}_2}$.

A symmetric generalization of (18) for splitting into three or more vector fields is also possible. Consider a splitting into k vector fields to obtain the symmetric method:

$$\varphi_{\mathbf{f}_1,1/2} \circ \cdots \circ \varphi_{\mathbf{f}_{k-1},1/2} \circ \varphi_{\mathbf{f}_k,1} \circ \varphi_{\mathbf{f}_{k-1},1/2} \circ \cdots \circ \varphi_{\mathbf{f}_1,1/2}\,. \tag{21}$$

An s-stage composition method with (21) as a submethod requires $s \times (2k-1)$ evaluations per step. Implemented directly as a splitting method, composing flows as in (19) and using the FSAL property reduces the cost to $s \times 2\,(k-1)$ evaluations per step. All experiments in the sequel use this technique to reduce the amount of work involved which, unless otherwise stated, is estimated by counting the number of evaluations of $\mathbf{f}_i$, $i = 1, \ldots, k$.

5.1 *Kepler problem*

Kepler's problem describes the motion in the configuration plane of a material point that is attracted towards the origin with a force inversely proportional to the square of the distance. In non-dimensional form, the Hamiltonian is:

$$H(\mathbf{p}, \mathbf{q}) = \frac{1}{2}\,(p_1^2 + p_2^2) - \frac{1}{\sqrt{q_1^2 + q_2^2}}.$$

The corresponding differential system is:

$$\mathbf{p}' = -\frac{\partial H}{\partial \mathbf{q}}, \quad \mathbf{q}' = \frac{\partial H}{\partial \mathbf{p}}\,. \tag{22}$$

The method (20) can be seen to correspond to a symplectic partitioned Runge–Kutta method (see for example [21]) by defining

$$\mathbf{f}_1 = \begin{pmatrix} \mathbf{0} \\ \dfrac{\partial H}{\partial \mathbf{p}} \end{pmatrix}, \quad \mathbf{f}_2 = \begin{pmatrix} \dfrac{-\partial H}{\partial \mathbf{q}} \\ \mathbf{0} \end{pmatrix}.$$

Integrating the two separate systems (22), it follows that the exact flows $\varphi_{\mathbf{f}_1,h}$ and $\varphi_{\mathbf{f}_2,h}$ coincide with explicit Euler steps [1].

Consider initial conditions $p_1(0) = 0$, $p_2(0) = \sqrt{(1+e)/(1-e)}$, $q_1(0) = 1 - e$, $q_2(0) = 0$, with eccentricity $e = 3/5$ and an integration interval of $[0, 15/2]$. The orbit has period 2π. The high precision reference solution is:

$$\begin{aligned}
p_1(15/2) &= -0.856384715343395351524486215029 51, \\
p_2(15/2) &= -0.160552150799838435254419104101 90, \\
q_1(15/2) &= -0.828164402690770818204757585370 33, \\
q_2(15/2) &= 0.778898095658635447081654480795 37.
\end{aligned}$$

Step sizes are selected from $3/2^i$, $i = 3, 7/2, 4, \ldots, 10$, and the work is estimated by counting the number of force evaluations $\mathbf{f}_2$ which are dominant.

5.2 *Lorenz equations*

In his original article, on numerical weather prediction, Lorenz gave a simple three dimensional nonlinear system with parameters $\sigma = 10$, $r = 28$ and $b = 8/3$ for which trajectories produce

a strange attractor [22]:

$$y_1' = \sigma(y_2 - y_1), \quad y_2' = ry_1 - y_2 - y_1 y_3, \quad y_3' = -by_3 + y_1 y_2. \tag{23}$$

Consider splitting the right hand side of (23) into nonlinear and linear parts $\mathbf{f}_1$ and $\mathbf{f}_2$, respectively (see for example [23]):

$$\begin{aligned} &y_1' = 0, && y_2' = -y_1 y_3, && y_3' = y_1 y_2, \\ &y_1' = \sigma(y_2 - y_1), && y_2' = ry_1 - y_2, && y_3' = -by_3, \end{aligned} \tag{24}$$

with exact solutions $\varphi_{\mathbf{f}_1,h}$ and $\varphi_{\mathbf{f}_2,h}$. The Lorenz equations (23) are an example of a conformal volume preserving system, where the divergence of the vector field is a constant. An interesting feature of splitting methods based on (24) is that they are conformal volume preserving integrators.

Consider initial conditions $y_1(0) = 10$, $y_2(0) = -20$, $y_3(0) = 20$ with an integration interval of [0, 1]. The high precision reference solution is:

$$\begin{aligned} y_1(1) &= 8.635692709892506017930544628 6390, \\ y_2(1) &= 2.798663387927457052023080059 0652, \\ y_3(1) &= 33.360635089731421577891858462673. \end{aligned}$$

Step sizes are selected from 2^{-i}, $i = 3, 7/2, 4, \ldots, 9$.

5.3 *Euler equations*

The motion of a rigid body, whose centre of mass is fixed at the origin, is modelled by Euler's equations [24]:

$$y_1' = \frac{I_2 - I_3}{I_2 I_3} y_2 y_3, \quad y_2' = \frac{I_3 - I_1}{I_3 I_1} y_3 y_1, \quad y_3' = \frac{I_1 - I_2}{I_1 I_2} y_1 y_2. \tag{25}$$

The vector $\mathbf{y} = (y_1, y_2, y_3)^{\mathrm{T}}$ represents the angular momentum in the body frame, while I_1, I_2, I_3 are the principal moments of inertia. The system (25) has two quadratic first integrals:

$$\begin{aligned} H(\mathbf{y}) &= \frac{1}{2}\left(\frac{y_1^2}{I_1} + \frac{y_2^2}{I_2} + \frac{y_3^2}{I_3}\right), \\ I(\mathbf{y}) &= y_1^2 + y_2^2 + y_3^2. \end{aligned} \tag{26}$$

The initial conditions are taken to be $y_1 = \cos(11/10)$, $y_2 = 0$, $y_3 = \sin(11/10)$. The values of the principal moments are $I_1 = 2$, $I_2 = 1$, $I_3 = 2/3$. The integration interval is [0, 200]. With these initial values, the invariant $I(\mathbf{y})$ has the effect of constraining the motion from $\mathbb{R}^3$ to the unit sphere. The invariant $H(\mathbf{y})$ is an ellipsoid and, in conjunction with $I(\mathbf{y})$, has the effect of constraining the motion to an ellipse on the unit sphere.

The differential system (25) can be split into three components, H_1, H_2 and H_3, each of which is Hamiltonian:

$$\begin{aligned} &H_1\colon y_1' = 0, && y_2' = \frac{y_1 y_3}{I_1}, && y_3' = -\frac{y_1 y_2}{I_1}, \\ &H_2\colon y_1' = -\frac{y_2 y_3}{I_2}, && y_2' = 0, && y_3' = \frac{y_1 y_2}{I_2}, \\ &H_3\colon y_1' = \frac{y_2 y_3}{I_3}, && y_2' = -\frac{y_1 y_3}{I_3}, && y_3' = 0, \end{aligned} \tag{27}$$

with exact solutions $\varphi_{\mathbf{f}_1,h}$, $\varphi_{\mathbf{f}_2,h}$, $\varphi_{\mathbf{f}_3,h}$. The splitting (27) was derived independently by McLachlan [25] and Reich [26].

The high precision reference solution is:

$$\begin{aligned} y_1(200) &= -0.2832477351486134619661053543504 8, \\ y_2(200) &= 0.5010392437833748521853580165465 5, \\ y_3(200) &= 0.8177593758081681490217759921257 3. \end{aligned}$$

Step sizes are selected from 2^{-i}, $i = 0, 1/2, \ldots, 5$.

Note that the system (25) is a Lie–Poisson system. Since $I(\mathbf{y})$ is a Casimir, it is conserved exactly by the splitting method (a composition of Poisson maps preserves Casimirs). The error in the invariant $H(\mathbf{y})$ remains bounded but it is not conserved exactly. For an example of a method that preserves both invariants see ref. [1, chapter VII].

5.4 *Generalized Arter flow*

Source-free vector fields are of particular interest in fluid dynamics. The Arter flow, a trigonometric flow involving sin, is one such example. Volume preserving integrators are particularly advantageous for source-free vector fields. Several splitting methods based on implicit integrators are known (see for example [27]). A splitting procedure based on the composition of *explicit* flows that works for polynomials of trigonometric functions is presented in ref. [28]. Each flow can be computed using the explicit Euler method. In this section, we consider a four-dimensional generalization of the Arter flow [see ref. 28].

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(\mathbf{y}) = \sum_{i=1}^{7} \mathbf{f}_i(\mathbf{y}), \tag{28}$$

where

$$\begin{aligned} &\mathbf{f}_1 = c_1 \sin(y_2 + y_3) \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} \quad \mathbf{f}_2 = c_2 \sin(y_1 + y_2) \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \\ &\mathbf{f}_3 = c_3 \sin(y_1 - y_4) \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} \quad \mathbf{f}_4 = c_4 \cos(y_3 + y_4) \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ &\mathbf{f}_5 = c_5 \sin(y_1 - y_3) \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{f}_6 = c_6 \cos(y_1 + y_2) \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \\ &\mathbf{f}_7 = c_7 \sin(y_2 + y_3) \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned} \tag{29}$$

Each of the flows is divergence-free so that the composition yields a volume conserving integrator. The initial conditions are taken to be $y_1(0) = y_2(0) = y_3(0) = y_4(0) = 1/10$, with parameter values $c_1 = c_2 = -1/10$, $c_3 = c_4 = c_5 = c_6 = c_7 = 1/10$. The integration interval

is taken to be [0, 200]. The high precision reference solution is:

$$\begin{aligned} y_1(200) &= -2.532278630661324877888233884688 2, \\ y_2(200) &= 8.073666851556034964899539686 1803, \\ y_3(200) &= -6.797394842336377210794749423957 2, \\ y_4(200) &= -3.863353349650527654766721957233 4. \end{aligned}$$

The step sizes are selected from 2^{-i}, $i = -1, -1/2, \ldots, 4$.

5.5 *Comments on the results*

Figures 3–5 illustrate the results of various composition schemes applied to the problems of this section. For the methods of order six and order eight taking additional stages give some improvement over known methods. The method (12) performs best among the methods of order six, although the method (11) (which uses less stages) also performs well. For order eight, the method (13) performs the best overall and the additional stages used by the method (14) are not always advantageous.

All the new methods of order ten perform better than the known prior methods. Somewhat surprisingly, this includes the method (15) which uses less stages than previously known methods. The additional work in extra stages is consistently offset by increased accuracy. It was previously thought that such high order methods are not very competitive for computations in double precision arithmetic [1]. However, the best of the new tenth order methods (17) now outperforms the best eighth order method at the boundary with single precision arithmetic, which is a very substantial improvement.

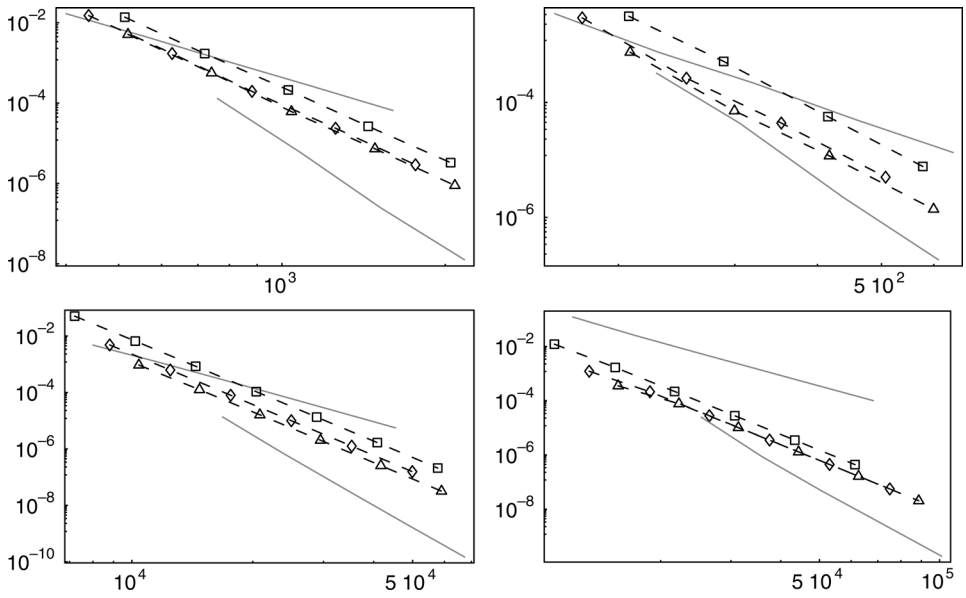

Figure 3. Work-precision comparison using a log–log scale for the Kepler problem (top left), the Lorenz equations (top right), Euler's equations (bottom left) and the generalized Arter flow (bottom right). Methods of order six: □ represents s9odr6b of Kahan and Li [10], ◇ represents (11) and △ represents method (12). Lines without symbols indicate the best methods of neighboring orders.

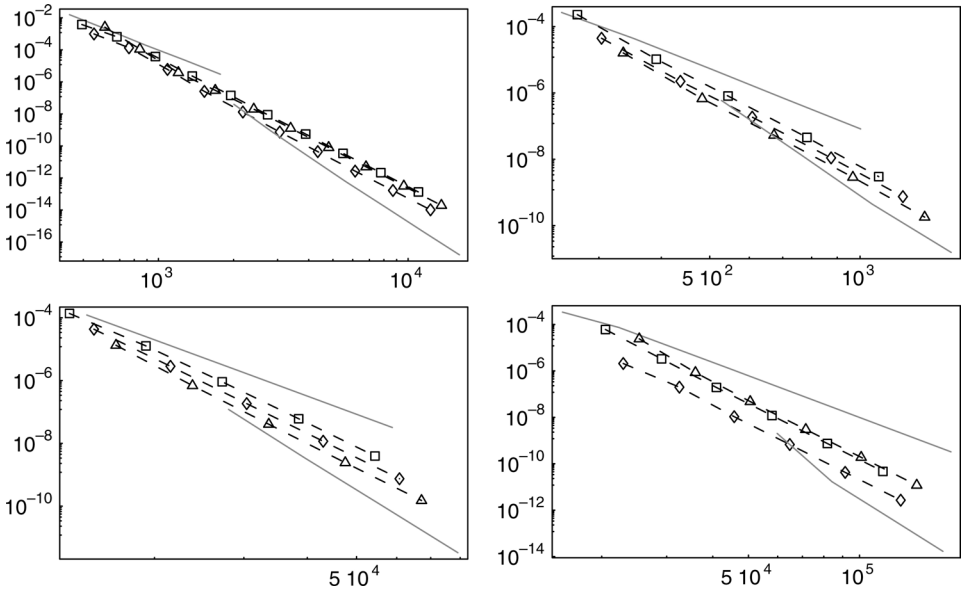

Figure 4. Work-precision comparison using a log–log scale for the Kepler problem (top left), the Lorenz equations (top right), Euler's equations (bottom left) and the generalized Arter flow (bottom right). Methods of order eight: □ represents s17odr8b of Kahan and Li [10], ◇ represents method (13) and △ represents method (14). Lines without symbols indicate the best methods of neighboring orders.

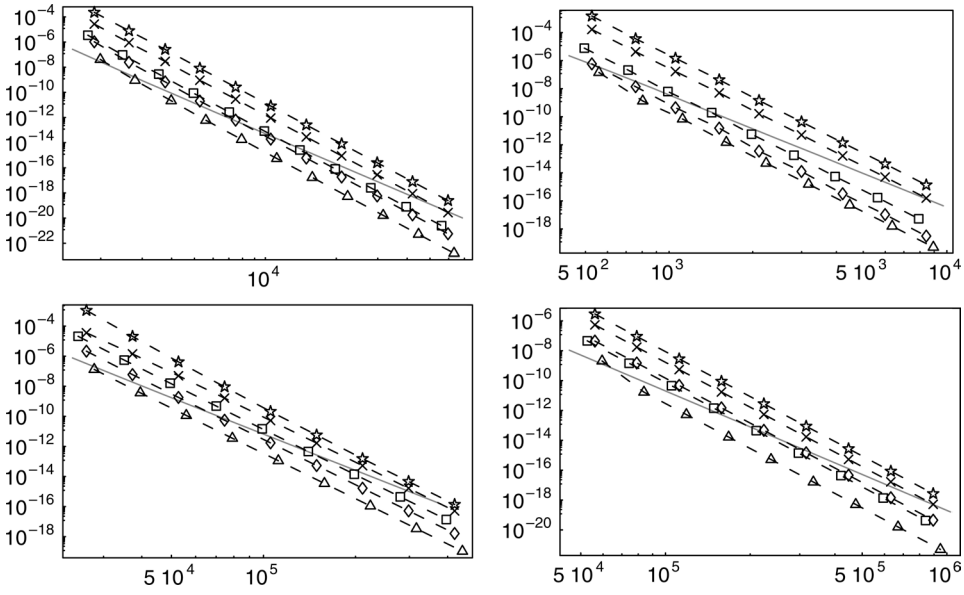

Figure 5. Work-precision comparison using a log–log scale for the Kepler problem (top left), the Lorenz equations (top right), Euler's equations (bottom left) and the generalized Arter flow (bottom right). Methods of order ten: ✰ represents s33odr10c of Kahan and Li [10], × represents (3.15) of Hairer *et al.* [1], □ represents method (15), ◇ represents method (16) and △ represents method (17). Lines without symbols indicate the best methods of neighboring orders.

6. Summary

The derivation of composition methods, particularly at high order, represents a very challenging optimization problem. We have presented results of an intensive numerical search and developed a unified framework for analyzing their qualitative properties. Using the framework, we have shown that the coefficients of order six and eight in ref. [10] appear to be optimal for the number of stages and we have presented alternatives which use additional stages to obtain a measurable improvement. In particular, methods (11) and (13) represent a reasonable compromise between improved accuracy and additional work. Several new tenth order composition schemes have also been derived and numerical experiments demonstrate significant improvements over previously known methods. In particular, the method (17) was found to be the most efficient at high accuracy. The numerical experiments also appear to indicate that there is room for composition schemes of order twelve in double precision arithmetic.

The numerical search for all of the methods of order six and eight terminated with the same minimal solution on each processor, which is very strong evidence that these are indeed global minima in the given search region. The same cannot be said of the methods of order ten, where only four to six processors had the same final solution, and it is still an open problem as to whether better methods exist. A selection of the best methods presented here have been built into the numerical differential equation solver in *Mathematica*.

The sign structure of composition coefficients has recently been investigated and could be used to improve the starting data for the numerical search [29]. It would also be interesting to investigate the use of low discrepancy points rather than randomly distributed points in the starting data. The approach outlined here could also be adapted to the derivation of symmetric composition of nonsymmetric methods, which are found in a number of practical applications.

Acknowledgements

The authors would like to thank P. C. Moan for fruitful discussions, R. Knapp for his work on numerical optimization in *Mathematica* and the organisers of the workshop 'Numerical Methods for Local and Global Optimization', Cortona, Italy, at which this work was presented. Thanks also to Apple Computer for providing the Xserve machine that was used for carrying out our numerical search and to the referees for constructive comments.

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